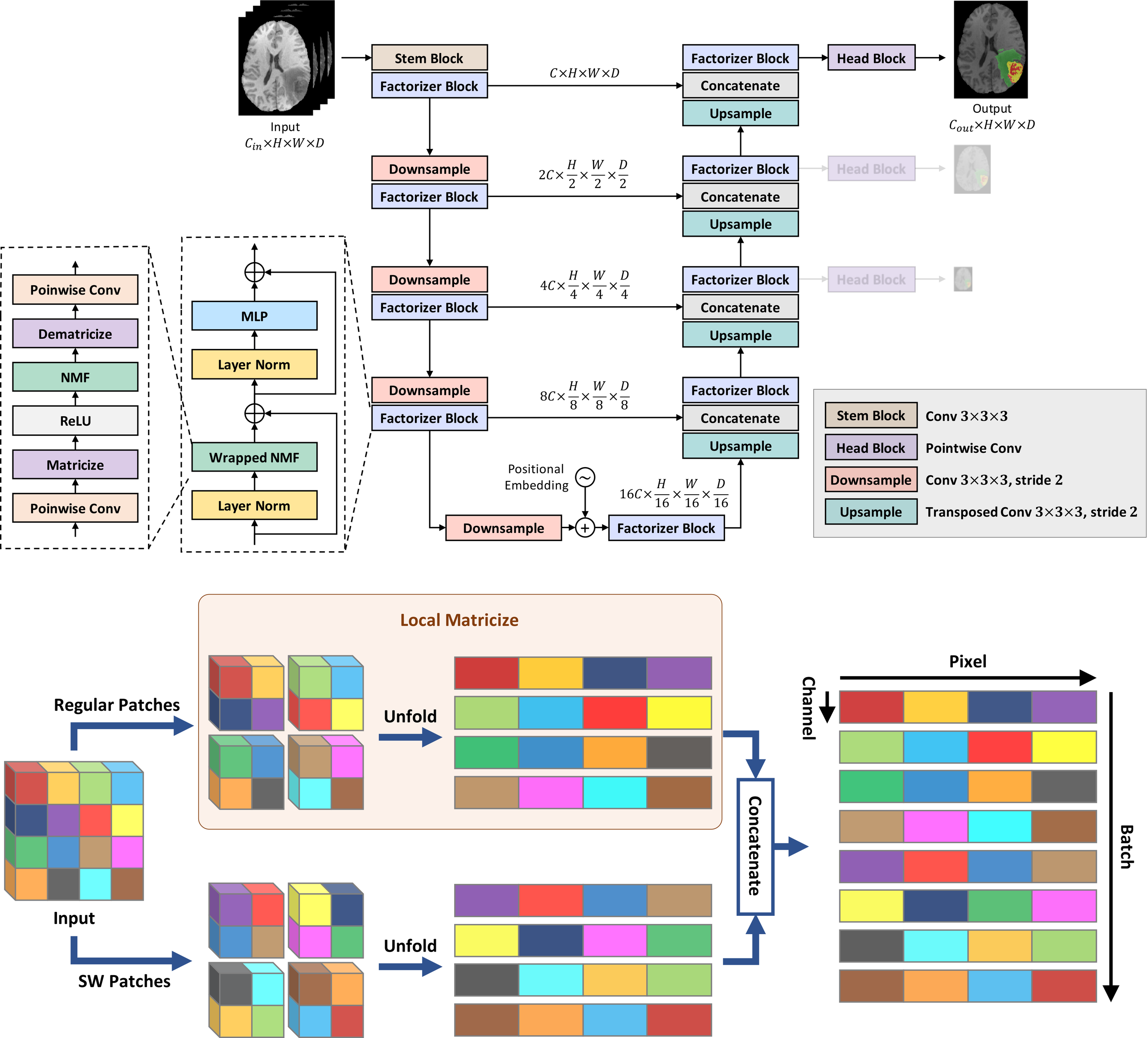
Input
$C_{in}\times H\times W\times D$
Stem Block
Factorizer Block
$C\times H\times W\times D$
Downsample
Factorizer Block
$2C\times\frac{H}{2}\times\frac{W}{2}\times\frac{D}{2}$
Downsample
Factorizer Block
$4C\times\frac{H}{4}\times\frac{W}{4}\times\frac{D}{4}$
Downsample
Factorizer Block
$8C\times\frac{H}{8}\times\frac{W}{8}\times\frac{D}{8}$
Positional Embedding
Downsample
$16C\times\frac{H}{16}\times\frac{W}{16}\times\frac{D}{16}$
Factorizer Block
Factorizer Block
Concatenate
Upsample
Factorizer Block
Concatenate
Upsample
Factorizer Block
Concatenate
Upsample
Factorizer Block
Concatenate
Upsample
Head Block
Head Block
Head Block
Output
$C_{out}\times H\times W\times D$
Poinwise Conv
Dematricize
NMF
ReLU
Matricize
Poinwise Conv
MLP
Layer Norm
Wrapped NMF
Layer Norm
Stem Block
Conv 3×3×3
Head Block
Pointwise Conv
Downsample
Conv 3×3×3, stride 2
Upsample
Transposed Conv 3×3×3, stride 2
Local Matricize
Regular Patches
Input
SW Patches
Unfold
Unfold
Concatenate
Pixel
Channel
Batch